Attention Module Based on Feature Normalization

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***Abstract*—Attention mechanism is an essential component in convolutional neural networks. Although attention modules like SE and CBAM have achieved tremendous success in fields such as image classification and object detection, most existing attention modules can only perform feature fusion in either channel or spatial dimensions. This greatly limits the flexibility of attention mechanisms to adapt in both channel and spatial dimensions, leading to underutilization of feature information. To address these issues, we propose a Feature Normalization-based Attention Module (FNAM) in this paper. This module generates a feature normalization weight map from the input feature map. During the normalization process, the scale and shift parameters for each channel are derived from the mean and standard deviation of the feature map. Consequently, this approach effectively utilizes feature information from all dimensions. To demonstrate the feasibility and effectiveness of this attention module, we conducted ablation experiments. In the image classification task, the FNAM module shows a significant improvement in the performance of classification networks on the CIFAR dataset compared to other mainstream attention modules. In the object detection task on the VOC dataset, the FNAM module outperforms most other attention modules. Particularly, concerning small and medium-sized objects in the dataset, its detection accuracy reaches the highest, with improvements of 1.6% and 1.5% respectively.**

Keywords: computer vision; feature similarity; feature normalization; attention module; CNNs

I. INTRODUCTION

Plug-and-play attention mechanisms are one of the hot research directions in the field of computer vision, and appropriate attention modules [1] can effectively improve the performance of deep neural networks. In previous literature [2], the significance of attention mechanisms has been widely investigated, and researchers believe that the core of attention mechanisms lies in enhancing the model's feature representation capability.

The currently common attention mechanisms in use are channel attention, channel and spatial mixed attention, and self-attention. The SE module [3] is one of the most representative channel attention modules, characterized by its simple structure, which allows adaptive adjustment of feature responses between channels through feature recalibration. This enables the enhancement of important feature maps and suppression of less important ones. The CBAM module [4], on the other hand, represents mixed attention by combining channel attention with spatial attention. It refines features along both the channel and spatial dimensions, providing a more comprehensive attention mechanism. The Non-local network module [5] was the first to introduce the idea of self-attention into the field of computer vision. It performs global context modeling on feature maps, effectively capturing long-range feature dependencies. By leveraging the global feature correlations, it achieves the enhancement and suppression of corresponding features. The currently popular Transformer [11] architecture is a typical representative of self-attention mechanisms, which has achieved outstanding results in various domains such as digital image processing and natural language processing.

However, the above attention mechanisms all have certain limitations, which restrict their performance improvement. Channel attention mechanism primarily relies on globally pooled features to compute attention weights for each channel, without utilizing feature information in the spatial domain. The channel and spatial mixed attention mechanism does not directly calculate a 3D attention weight map; instead, it separately computes channel attention weights and spatial attention weights, failing to fully utilize feature information from all dimensions. Self-attention mechanism can make full use of feature information in both channel and spatial domains, aggregating and processing global feature information. However, its large number of model parameters and heavy computational burden severely limit its development space in practical applications.

To better utilize feature information in both the channel and spatial domains, we propose a Feature Normalization-based Attention Module (FNAM) in this paper. This module generates a feature normalization weight map from the input feature map. During the normalization process, the scale transformation parameters and shift parameters for each channel are derived from the mean and standard deviation of the feature map. As a result, we can make more comprehensive use of feature information from all dimensions.

This paper verifies the performance of the proposed module through experiments. By observing the experimental data, it can be found that the FNAM module can achieve prediction accuracy comparable to that of the CBAM module in image classification tasks and object value estimation tasks with the same number of parameters as SE.

The main contributions of this paper are as follows:

1) We propose a lightweight attention module based on feature normalization.

2) We introduce a scale transformation and offset method based on the mean and standard deviation of the feature map.

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II. ALGORITHM

A. Feature Normalization

To overcome the problem of internal covariate shift during the training process of deep neural networks, Ioffe et al. [6] proposed the method of Batch Normalization. Let X be a feature map generated by a deep neural network, with dimensions of B×C×H×W, where B represents the batch size, C represents the number of channels, H represents the length of the feature map, and W represents the width of the feature map. Batch Normalization consists of two steps:

1) Normalize the feature map along the channels. Let x_{ijkl} be the feature value at the address index (i,j,k,l) in the feature map. Using (1) and (2), we can calculate the batch mean μ_j and standard deviation σ_j of the data (where ε is a sufficiently small positive number to avoid division by zero issues). The normalization is achieved using (3).

$$\mu_j = \frac{1}{BHW}\sum_{i=0}^{B-1}\sum_{k=0}^{H-1}\sum_{l=0}^{W-1} x_{ijkl} \tag{1}$$

$$\sigma_j = \sqrt{\frac{1}{BHW}\sum_{i=0}^{B-1}\sum_{k=0}^{H-1}\sum_{l=0}^{W-1}\left(x_{ijkl}-\mu_j\right)^2 + \varepsilon} \tag{2}$$

$$\hat{x}_{ijkl} = \frac{x_{ijkl}-u_j}{\sigma_j} \tag{3}$$

2) Scale transformation and offset. Because after the data is normalized, most of it will fall within the linear range of the activation function, this phenomenon severely limits the non-linear expressive capability of the neural network. Therefore, Batch Normalization introduces learnable parameters γ and β to perform scale transformation and offset on the normalized data using (4).

$$\tilde{x}_{ijkl} = \gamma_j \hat{x}_{ijkl} + \beta_j \tag{4}$$

This paper introduces the concept of feature normalization into the attention mechanism. The key difference is that, in this paper, feature normalization is applied to individual channels rather than batches.

$$\mu_{ij} = \frac{1}{HW}\sum_{k=0}^{H-1}\sum_{l=0}^{W-1} x_{ijkl} \tag{5}$$

$$\sigma_{ij} = \sqrt{\frac{1}{HW}\sum_{k=0}^{H-1}\sum_{l=0}^{W-1}\left(x_{ijkl}-\mu_{ij}\right)^2 + \varepsilon} \tag{6}$$

$$\hat{x}_{ijkl} = \frac{x_{ijkl}-u_{ij}}{\sigma_{ij}} \tag{7}$$

Additionally, it should be noted that the normalized scale transformation parameters and offset parameters will be dynamically adjusted with the mean and standard deviation of each channel feature map.

B. Feature Normalization-based Attention Module(FNAM)

As illustrated in Fig. 1, A brief introduction to the algorithmic flow of FNAM is as follows: First, the scale transformation factors are generated using the variance of the feature map. Next, the offset values are generated using the mean of the feature map. Then, these generated parameters are used to perform scale transformation and offset on the normalized feature map. Finally, the Sigmoid function is applied to constrain the attention weight values between 0 and 1.

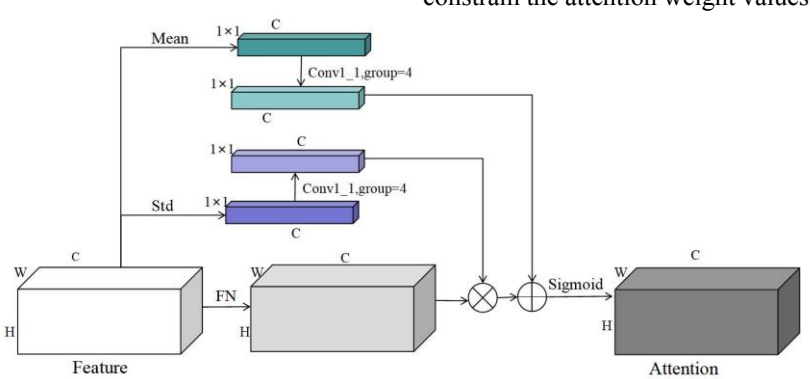

Fig. 1. The structure diagram of FNAM

In Fig. 1, "Mean" represents the global average pooling operation applied to the feature map, and "Std" represents the operation of computing the variance of the feature map. "FN" represents the normalization operation performed on the feature map along the spatial dimension.

Through the ECA [7] module, we observed that the operation of first grouping and then using a 1x1 convolution kernel can replace the fully connected layer operation. In Fig. 1, we use "Conv1_1, group=4" to represent this operation. This operation can effectively reduce computational complexity while still capturing long-range dependencies between channels, ensuring the comprehensive utilization of channel information in the feature map. Experimental results show that the idea of grouping is feasible and effective for FNAM modules.

III. EXPERIMENT

This paper conducts a series of ablation experiments on open-source datasets to validate the effectiveness of the proposed attention module. All network models and attention modules used for evaluation are implemented and executed on the PyTorch framework platform. Furthermore, to ensure the validity of the experimental results, the hyperparameter settings

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for different attention modules using the same backbone network are kept consistent.

A. Image classification task.

To demonstrate the effectiveness of the proposed attention module, this paper conducts image classification tasks on the CIFAR dataset. The CIFAR dataset is divided into two different subsets, namely CIFAR10 and CIFAR100, with CIFAR10 containing 10 classes and CIFAR100 containing 100 classes. Both datasets consist of 50k training samples and 10k validation samples, all of which are 32×32 color images. To validate the effectiveness and feasibility of the proposed attention module, it is integrated into several widely used network architectures. In the image classification task, the attention module is added to the output layer of each residual block.

In this experiment, an RTX2080 GPU was used for accelerated computation. Each module was trained for 160 epochs, and the batch size for all models was set to 128. The optimization strategy used was stochastic gradient descent with Nesterov momentum. The values for Nesterov momentum and weight decay were set to 0.9 and 0.0005, respectively. The initial learning rate was set to 0.07. After 80 epochs, an exponential decay strategy was applied, where the learning rate was reduced to 0.95 of its original value every epoch. After 120 epochs, the learning rate was kept constant at 0.0001.

To avoid the randomness caused by model initialization, each method in the experiment was tested five times under the same conditions, and the average accuracy of each method was reported to ensure the credibility of the ablation experiment. For other attention modules cited in the ablation experiment, the hyperparameters were set to the default values provided by the original authors' publicly available code (this setting remained the same for subsequent ablation experiments in other visual tasks).

The experimental results on the CIFAR10 dataset(C10) and CIFAR100 dataset(C100) are shown in in Tab. I.

TABLE I. TOP-1 ACCURACY (%) OF DIFFERENT ATTENTION MODULES ON DIFFERENT BACKBONE NETWORKS ON CIFAR DATASETS

Method	ResNet-18[9]		ResNet-34[9]		ResNet-50[9]	
	C10	C100	C10	C100	C10	C100
Baseline	92.37	68.87	93.07	71.14	93.96	73.40
+SE [3]	92.72	69.98	93.62	71.78	94.57	74.04
+CBAM[4]	92.94	**70.21**	93.79	71.85	**94.88**	74.46
+SimAM [8]	92.67	69.03	93.28	71.39	94.07	73.87
+FNAM	**92.98**	70.13	**93.81**	**71.92**	94.73	**74.53**

On the CIFAR10 and CIFAR100 datasets, when the proposed attention module FSNAM is integrated with different backbone networks, it consistently improves the classification accuracy of the original networks. In the ablation experiments, compared with other mainstream attention modules, the proposed attention module in this paper achieves the highest accuracy on most backbone networks.

To further compare the performance of each attention module, this paper uses ResNet as the backbone network and provides the model complexity and inference speed of the convolutional neural networks after adding the attention modules. The results are shown in Tab. II.

TABLE II. MODEL COMPLEXITY AND SPEED

Method	ResNet-18[9]			ResNet-50[9]		
	Parameter	FLOPs	Inference Speed	Parameter	FLOPs	Inference Speed
Baseline	11.18 M	37.22 M	439 FPS	23.52 M	84.34 M	210 FPS
+SE [3]	11.27 M	37.32 M	310 FPS	26.04 M	86.99 M	139 FPS
+CBAM [4]	11.27 M	37.41 M	182 FPS	26.07 M	89.40 M	77 FPS
+SimAM [8]	11.18 M	37.22 M	329 FPS	23.52 M	84.34 M	142 FPS
+FNAM	11.27 M	37.32 M	288 FPS	23.04 M	86.99 M	131 FPS

From Tab. II, it can be observed that the classification network with the FNAM module has parameter and FLOPs comparable to SE. The inference speed is slower than SE but still better than CBAM.

In addition, by observing the data in Tab. I and Tab. II, it can be found that in the classification task, the proposed attention module in this paper can achieve detection accuracy comparable to that of the CBAM module with the same number of parameters as SE.

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B. *Object detection task.*

This paper conducts the object detection task on the VOC dataset, which contains 20 classes. The VOC07+12 dataset combination is used, where the model is first trained on the trainval dataset of VOC2007 and VOC2012 (a total of 16,551 images) and then tested on the test dataset of VOC2007 (a total of 4,952 images). The experiments are performed using an RTX3080 GPU for accelerated computation, and each module is trained for 300 epochs with a batch size of 32.

All models in the experiments are pretrained on the ImageNet1K dataset and then fine-tuned on the VOC dataset. The optimization strategy used is stochastic gradient descent with Nesterov momentum. The values for Nesterov momentum and weight decay are set to 0.9 and 0.0005, respectively. The initial learning rate is set to 0.001, and after 70 epochs, an exponential decay strategy is applied, reducing the learning rate to 0.95 of its original value every epoch.

In the object detection task, this paper selects Yolov4-tiny [10] as the object detection network and adds attention modules to the output layer of the backbone network. All images are resized to 416×416. The experimental results on the VOC dataset are shown in Tab. III.

As shown in Tab. III, the FNAM module proposed in this paper outperforms the base network in all accuracy metrics. Additionally, by observing the AP_S, AP_M, and AP_L, it can be seen that the FNAM module shows the best improvement in detection accuracy for small and medium-sized objects. Through the visual results in Fig. 2, it can be observed that the proposed module helps the network capture more feature information of small and medium-sized objects.

TABLE III. ACCURACY (%) OF DIFFERENT ATTENTION MODULES ON VOC DATASETS ON YOLOV4-TINY

Method	AP	AP_{50}	AP_S	AP_M	AP_L
Baseline	34.0	66.7	9.0	24.3	38.1
+SE [3]	34.8	68.6	10.3	25.5	38.5
+CBAM [4]	**35.3**	68.9	9.8	25.1	**39.7**
+SimAM [8]	34.3	67.7	8.9	24.7	38.3
+FNAM	35.1	**69.7**	**10.6**	**25.8**	39.1

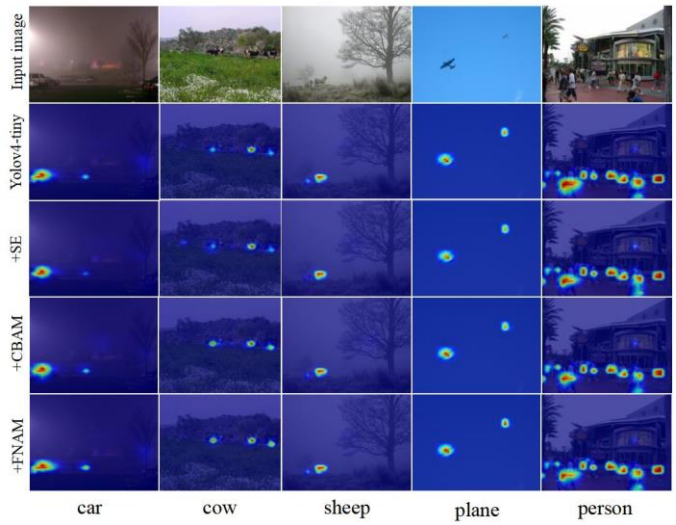

Fig. 2. The structure diagram of FNAM

IV. CONCLUSION

This paper proposes a feature normalization-based Convolutional Neural Network Attention Module (FNAM) that effectively utilizes information from all dimensions of the feature map to generate a corresponding 3D attention weight. The attention module first calculates the scale transformation factor using the feature map's variance and generates the offset factor using its mean. Then, it applies these generated parameters to normalize the feature map by performing scale transformation and offset. Finally, it generates the corresponding attention weights. The paper conducts ablation experiments using the PyTorch framework, and the results demonstrate that using the proposed attention module significantly improves the performance of the base network in image classification and object detection tasks.

Comparing with other attention modules, it can be observed that the FNAM module is very close to the SE module in terms of parameter count and inference speed. In addition, it can be seen from Fig. 1 that the structure and implementation process of the FNAM module are very simple, easy to reproduce in code, and very convenient to use. However, in terms of classification accuracy and object detection evaluation metrics, FNAM performs better than SE.It is worth noting that this paper uses the mean and standard deviation of each channel to utilize spatial feature information, which is a global spatial feature information. In contrast, CBAM uses convolution kernels to process spatial feature information within the receptive field. FNAM still lacks in utilizing local spatial feature information, especially for local spatial feature information. How to further utilize local spatial feature information is a direction for improving FNAM.

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REFERENCES

[1] Dai T, Cai J, Zhang Y, et al. Second-order attention network for single image super-resolution[C]//Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. 2019: 11065-11074.

[2] Mnih V, Heess N, Graves A. Recurrent models of visual attention[J]. Advances in neural information processing systems, 2014, 27.

[3] Hu J, Shen L, Sun G. Squeeze-and-excitation networks[C]//Proceedings of the IEEE conference on computer vision and pattern recognition. 2018: 7132-7141.

[4] Woo S, Park J, Lee J Y, et al. Cbam: Convolutional block attention module[C]//Proceedings of the European conference on computer vision (ECCV). 2018: 3-19.

[5] Wang X, Girshick R, Gupta A, et al. Non-local neural networks[C]//Proceedings of the IEEE conference on computer vision and pattern recognition. 2018: 7794-7803.

[6] Ioffe S, Szegedy C. Batch normalization: Accelerating deep network training by reducing internal covariate shift[C]//International conference on machine learning. pmlr, 2015: 448-456.

[7] Wang Q, Wu B, Zhu P, et al. ECA-Net: Efficient channel attention for deep convolutional neural networks[C]//Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. 2020: 11534-11542.

[8] Yang L, Zhang R Y, Li L, et al. Simam: A simple, parameter-free attention module for convolutional neural networks[C]//International conference on machine learning. PMLR, 2021: 11863-11874.

[9] He K, Zhang X, Ren S, et al. Deep residual learning for image recognition[C]//Proceedings of the IEEE conference on computer vision and pattern recognition. 2016: 770-778.

[10] Bochkovskiy A, Wang C Y, Liao H Y M. Yolov4: Optimal speed and accuracy of object detection[J]. arXiv preprint arXiv:2004.10934, 2020.

[11] Liu Z, Ning J, Cao Y, et al. Video swin transformer[C]//Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. 2022: 3202-3211.

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